



SIMPÓSIO BRASILEIRO DE **PESQUISA OPERACIONAL**



Optimization of Wagon Maintenance Mix Allocation in Freight Railways Using Integer Linear Programming

Otimização do Mix de Alocação da Manutenção de Vagões em Ferrovias de Carga por Programação Linear Inteira

Carmo Crediney Melo · José Cristiano Pereira · Luciano M. S. Varricchio · Prudente J. T. Aguiar · Felipe M. Lopes · Bruno M. O. Jordão

Catholic University of Petrópolis (UCP) · Poster Session · LVIII SBPO, Belo Horizonte, Sept 28 – Oct 1, 2026



Context and research questions



Why the problem matters

- Brazilian railways moved > **540 million tonnes** in 2024 (Ministry of Transport).
- Fleets of **~20,000 wagons** and ~18,000 interventions per year.
- Deciding **where** and **when** to maintain each wagon affects cost, fleet availability and workshop utilization.
- Light scopes (VR1, VR2, VMC) fit any workshop; heavy overhauls (VR4, VRG) only at **PM2**.
- Transfers add logistics and unavailability cost: sometimes **postponing is better**.
- Monthly capacity drops with vacations and equipment repairs.

Research questions

α

How to allocate maintenance demand across workshops, minimizing total cost without violating operational constraints?

β

How much does jointly handling capacity, technical eligibility and composite costs improve the plan over local rules or heuristics?

γ

How much does an optimization model raise the analytical quality and managerial gains of tactical planning?

The letters α , β and γ mark, in the next slides, where each question is answered.

Integer linear programming model



Variables: x_{gvetpm} = wagons of region g , type v , scope e , in month t , served at workshop pm ; q_{gvet} = backlog at month end.

Minimum total cost = service + backlog penalty

$$\min Z = \sum c \cdot x + \sum p \cdot q$$

Composite cost: maintenance + logistics + unavailability + retention

$$c = c_{\text{maint}} + c_{\text{log}} + c_{\text{unav}} + c_{\text{ret}}$$

Demand–backlog balance across months

$$\sum_{pm} x + q_t = d_t + q_{(t-1)}$$

Effective capacity in man-hours (min. and max.) per workshop and month

$$CapMin \leq \sum h \cdot x \leq CapMax$$

Technical eligibility and integrality

$$x \leq U \cdot a \quad x, q \in \mathbb{Z}_+$$

What the model decides

- How many wagons of each group go to each workshop, month by month.
- How many stay in backlog, paying a finite penalty.
- Respects minimum and maximum man-hour load and eligibility by scope.
- Separates technical infeasibility from economically justified postponement.

Implementation: Python + SciPy, HiGHS solver (MILP, gap 10^{-4}). Data in CSV.

Anonymized test instance



1,004

interventions

12

months

3

regions · 3 workshops

1,152

integer variables · 396 constraints

Scope	PM1	PM2	PM3	MH GDT	MH GDU
VR1 (light)	✓	✓	✓	8	9
VR2 (light)	✓	✓	✓	12	14
VR4 (heavy)	✗	✓	✗	30	35
VRG (heavy)	✗	✓	✗	40	46
VMC (corrective)	✓	✓	✓	10	12

Technical eligibility and man-hours per wagon. GDU difficulty index = $150/130 = 1.154$ (gross weight).

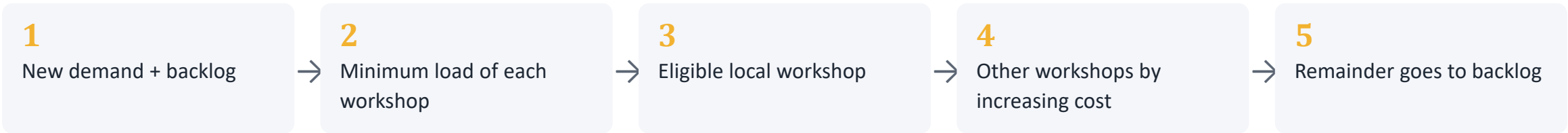
Aggregate slack, local bottleneck

- Annual load of **14,556 MH** against **15,664 MH** effective: 7.6 % slack.
- But heavy scopes require **6,593 MH** and PM2 only has **6,525 MH**.
- PM1 and PM3 have spare hours, but cannot perform VR4 or VRG.
- Vacations and repairs reduce capacity in specific months.

Fair comparison and ablation



Local-first heuristic under the same constraints as the model: demand, eligibility, minimum and maximum load, backlog.



Scenario	Total cost (MU)	Final backlog	Remote
Local-first heuristic	228,672	39	306
Full ILP	170,690	9	401
Direct cost only	178,447	9	612
No minimum load	170,591	9	398
LP relaxation	169,297	8.68	—

Both methods obey the same rules; they differ in decision quality, not in constraints.

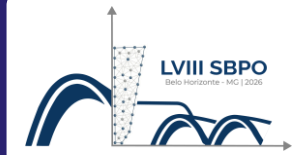
What the ablation shows

- Planning by direct cost alone **doubles the transfers** (401 → 612) and costs more in total.
- The minimum load is almost free: 0.06 % of the objective.
- The LP relaxation yields **fractional wagons** (8.68 in backlog); integrality gap 0.82 %.

Results: heuristic × optimized

β

γ



39 → 9

final backlog (wagons)
-76.9 %

-54.4 %

cumulative backlog
274 → 125 wagon-months

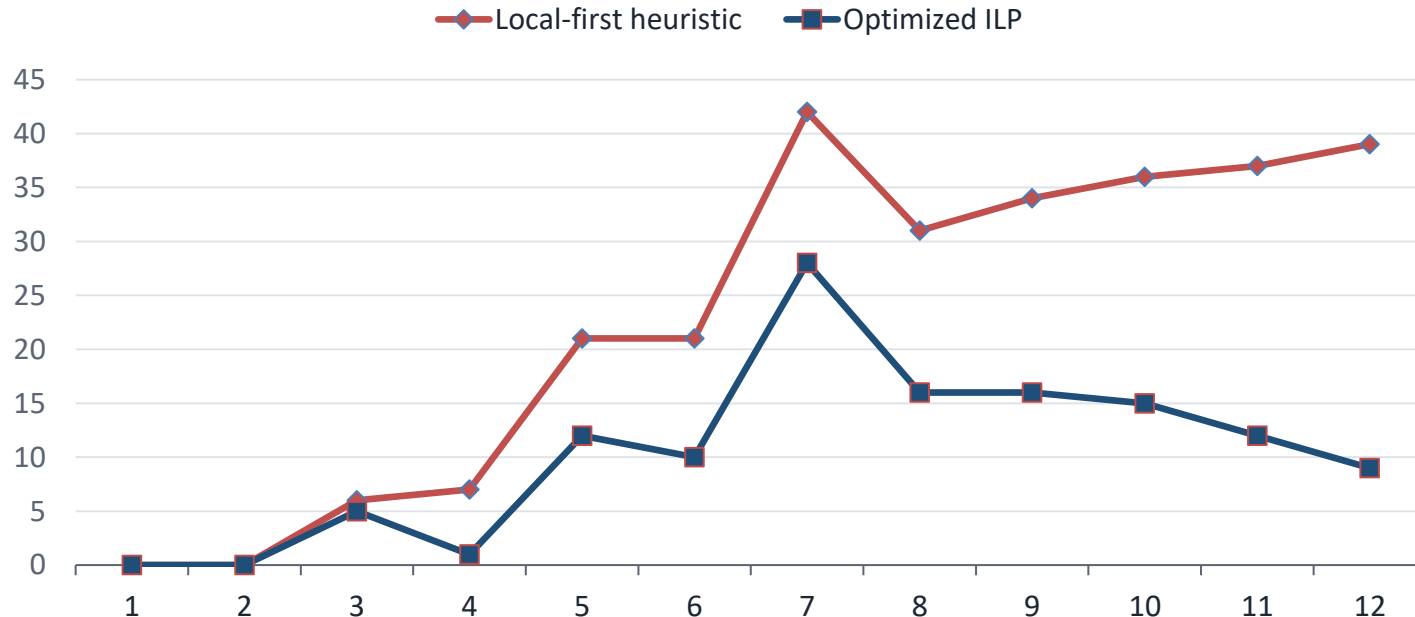
-25.4 %

total system cost
228.7k → 170.7k MU

< 1 s

solution time
11,520 variables in < 6 s

Wagons in backlog at the end of each month



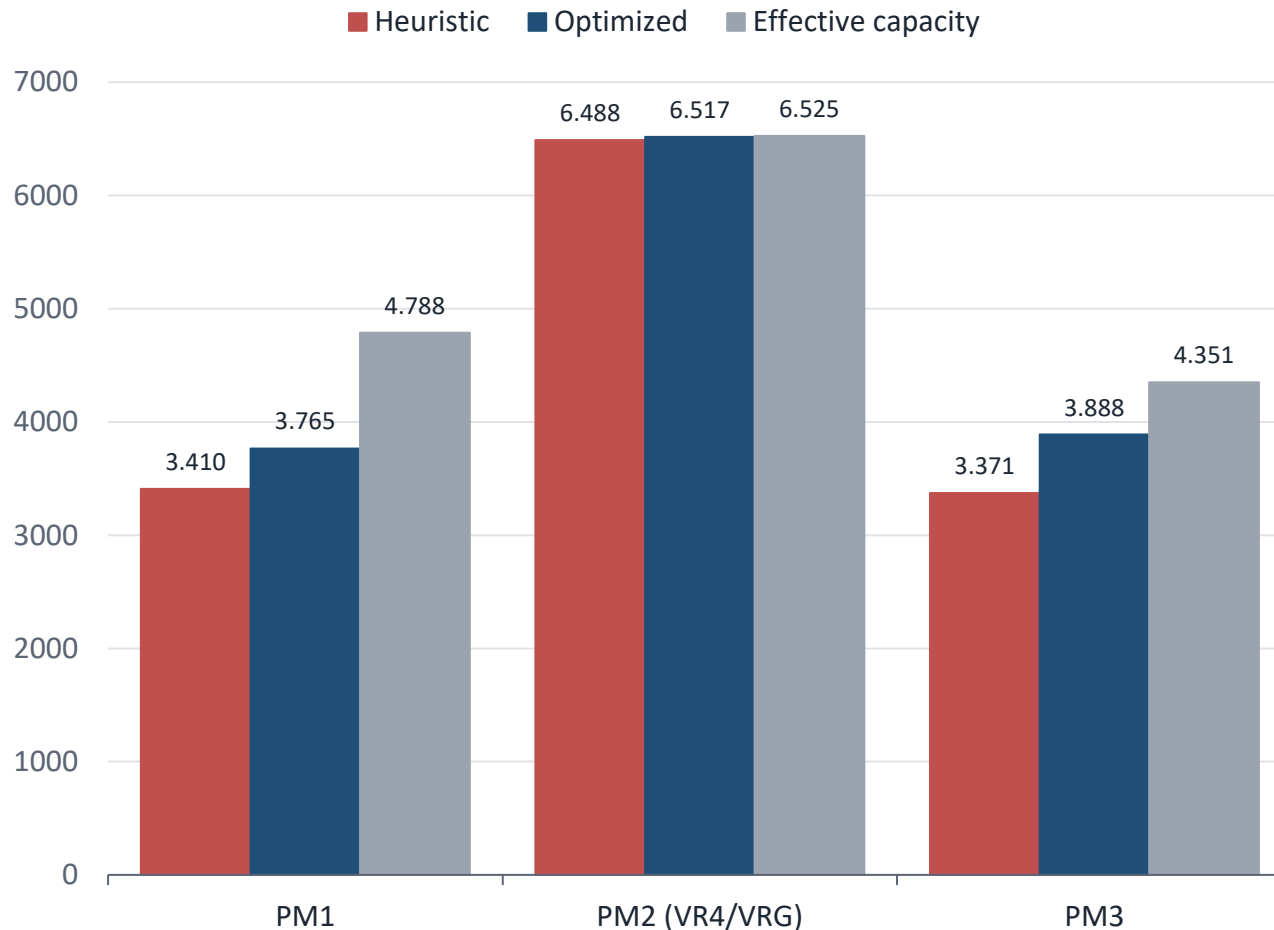
Reading

- Served **995** wagons versus **965**.
- The heuristic's queue grows from month 5 and never recovers; the optimum peaks in month 7 and falls.
- More remote moves (401 vs 306), but only when they **cost less than waiting**.

Source of the gain: the PM2 bottleneck



Annual man-hours per workshop



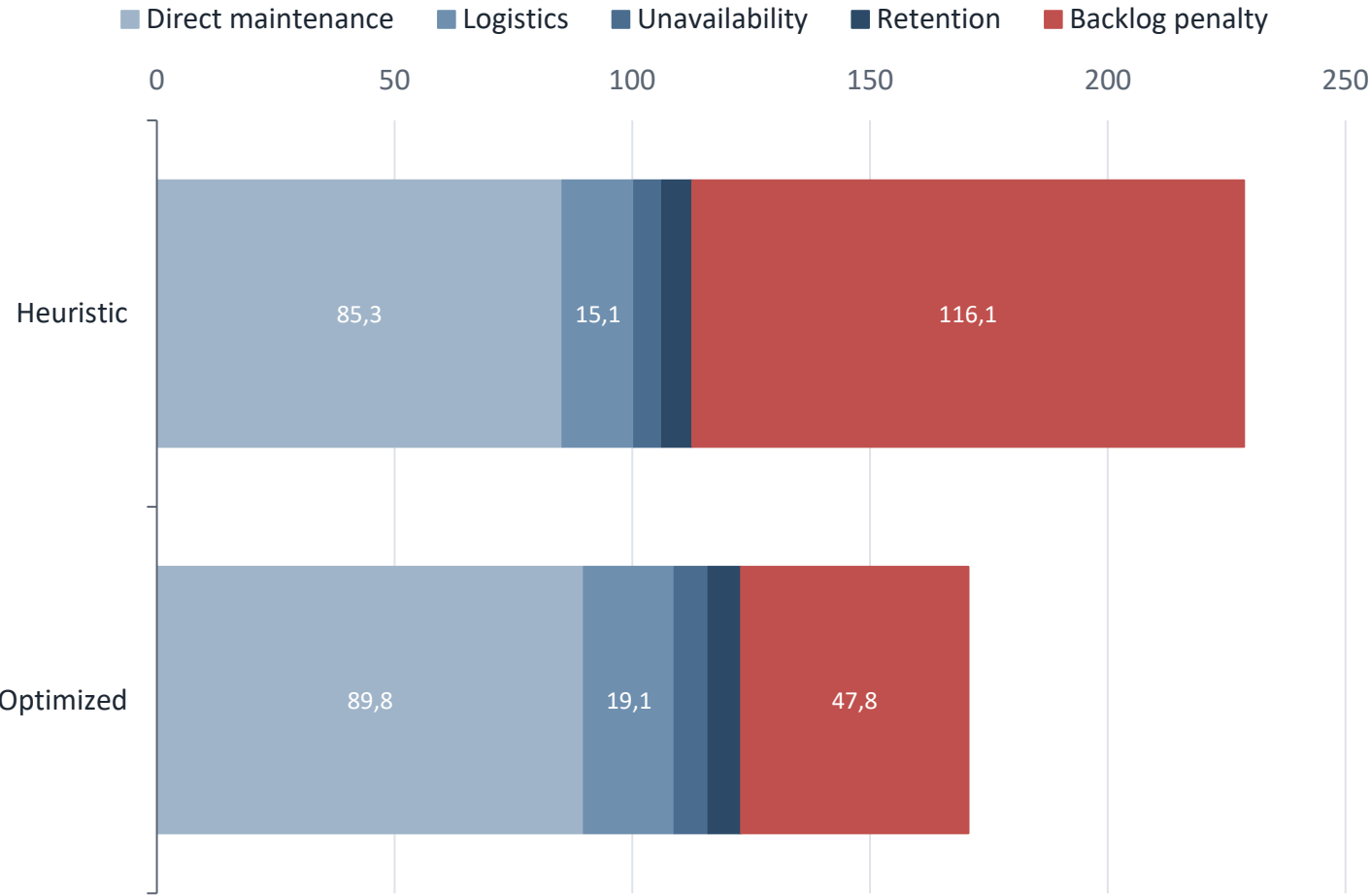
The mechanism

- PM2 is the only workshop that performs VR4 and VRG, and heavy demand already **exceeds** its capacity.
- The heuristic still sends **119 light wagons** to PM2 (103 VMC + 16 VR1), just because they belong to its region.
- Result: only **40 VR4 overhauls** in the year and 216 wagon-months of VR4 in queue.
- The optimum sends only **34 light wagons** to PM2, shifts light work to PM1 and PM3 and performs **75 VR4**.
- VR4 backlog drops from **216 to 41** wagon-months.

Cost: more transport, much less waiting



Components of total cost (thousand normalized MU)



+9 %

service cost
112.5k → 122.9k MU

-59 %

backlog penalty
116.1k → 47.8k MU

-25.4 %

total system cost
228.7k → 170.7k MU

Conclusions



α

The flow-balance model allocates service and backlog across months and separates technical infeasibility from economically justified postponement.

β

Against the local-first heuristic, under the same rules: 995 vs 965 wagons served, cumulative backlog -54.4 %, final backlog -76.9 %, total cost -25.4 %. The ablation confirms the role of composite cost, eligibility, minimum load and integrality.

γ

Solves in under 1 s (and 11,520 variables in under 6 s), which allows testing scenarios of downtime, capacity, outsourcing, equipment and transfers within the monthly tactical cycle.

Managerial lesson: aggregate man-hour slack does not guarantee service. Capacity is differentiated by scope, workshop and month, and that is where optimization finds the gain.

Next steps

- Demand uncertainty (scenarios and stochastic programming).
- Detailed network capacity and transfer routes.
- Backlog age and priority by criticality.
- MRO inventory availability and outsourcing.
- Validation with concessionaire data.



SIMPÓSIO BRASILEIRO DE **PESQUISA OPERACIONAL**



Thank you! · Obrigado!

Carmo Crediney Melo · carmo.42640028@ucp.br

Catholic University of Petrópolis (UCP)

Full paper in the SBPO 2026 proceedings: proceedings.science/sbpo-2026/trabalhos/optimization-of-wagon-maintenance-mix-allocation...

FAQ, poster (PT/EN), code and data: carmonexus.github.io/wagon-maintenance



Paper · SBPO
proceedings



FAQ · code · data



LinkedIn

Normalized monetary values; anonymized, reproducible instance.